

REFLOWECONOMY · REQUIREMENTS

ReflowEconomy requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-09-30	Amish Chadha	Design for construction (RFE-DDR-003): R3, R5 and R10 figures updated; R10 reported against its value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). R8 restated (process losses counted); new R16 intake quality rule; R10 value \$24,200 with the 1.5 kW fume fan; R11 fan 1.5 kW; shell and press assumptions updated
0.3	2026-09-25	Amish Chadha	Apply RFE-DDR-001 decisions (R2 redefined for the aluminium add-on bay, R13 and R14 name schema v0.2, single-phase supply assumption) and replace the TRL 2 status with the TRL 3 status from RFE-CAL-001
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (15 measurable requirements for the playbook, reference micro-factory and material passport, with status)
0.1	2026-09-24	Amish Chadha	Initial scaffold

ReflowEconomy requirements

These are first-pass requirements for the playbook, its reference micro-factory and the material passport. Targets are proposals for review, not user-validated needs, and will be revised after co-design sessions. Status is judged at TRL 3 against the calculation note RFE-CAL-001 v0.3 (docs/04-calcs/01-sizing.md , results in docs/04-calcs/results.csv): **3 are not met (R1, R4, R8)**, 2 are at risk (R6, R12), 2 cannot be verified at TRL 3 (R11, R15), 8 are met on paper, and 1 (R10) is \$300 over its value-engineering target. Decisions from RFE-DDR-001 (Amish, 2026-09-25) are applied: R2 is redefined for the aluminium add-on bay, and R13 and R14 name passport schema v0.2. Decisions from RFE-DDR-002 (recommendations accepted by Amish, 2026-09-25) are applied: R8 is restated to count process losses, and R16 adds an intake quality rule.

■ Playbook content

ID	Requirement	Target	Verification	Status at TRL 3
R1	Say which materials a small operation can recover safely	Feasibility matrix covers at least 7 material groups (plastics, aluminium, steel, glass, lithium cells, e-waste, ore), each with route, feasibility rating, main hazard and export point, and at least one cited source per row	Document review	Not met: the matrix has routes, ratings and reasons for 7 groups; hazard, export point and source columns are still to add
R2	Give a process recipe for each locally processed material	One recipe each for PET, HDPE, PP and aluminium: input grade, steps, temperature window, drying, yield, rejects and PPE, each sourced. Redefined by RFE-DDR-001 item 1: the aluminium recipe serves the add-on bay, which has its own safety case, not the reference floor	Document review against published processing data	Met: four sourced recipes in docs/playbook/recipes/
R3	Show whether a micro-factory can pay its way	Economics model per kilogram and per shift with local inputs (feedstock price, wage, energy tariff, product prices) and a break-even output; editable by an operator	Model review; later with at least one operating recycler	Met on paper: docs/playbook/economics_model.py with economics_inputs.csv ; result +\$0.73 per shift before rent, break-even product price \$2.48/kg, break-even input 99 kg per shift
R4	Export only what needs industrial refining	Every export stream names the refining step it needs, the licensed receiver type and the packing rule (for example, cells taped and in a fire-safe container)	Document review	Not met: principle and the cell packing rule only; per-stream rules still to write

■ Reference micro-factory

ID	Requirement	Target	Verification	Status at TRL 3
R5	Fit a small site	Floor area 60 m ² or less (two 40 ft container footprints or a small workshop), with aisles of 1.0 m or more and a separate hot zone	Layout drawing (RFE-DWG-001) and RFE-CAL-001 section 8	Met: 59.4 m ² , clear aisle 1.57 m, aisle headroom 2.04 m, hot zone 1.00 m from stock (at the limit)
R6	Process a useful amount per shift	100 kg or more of mixed collected input per 8 h shift	Time budget from machine ratings (RFE-CAL-001 section 3)	At risk: shredder needs 3.79 h of 4 h at an assumed 15 kg/h (published range 8.8 to 41.8 kg/h); sheet press has 2.06 sheets of capacity for the 2 needed
R7	Keep value local	50 % or more of input mass remanufactured or sold locally as clean feedstock	Mass balance, then site data	Met on paper: 51.6 %
R8	Limit what goes to disposal	Residue to licensed disposal 20 % or less of input mass, counting sorting residue and process losses together; nothing burned in the open. Restated by RFE-DDR-002 (item 8): the honest total is kept and R16 controls input quality	Mass balance, then site data	Not met for the reference input: 28.4 % (20 % sorting residue plus 8.4 % process losses). A load at the R16 threshold gives 19.5 %
R9	Use little energy	1.0 kWh or less per kilogram of output (products plus clean flake)	Energy budget per shift	Met on paper: 0.83 kWh/kg (42.9 kWh per shift)
R10	Stay affordable	Equipment for items 1 to 12 of <code>bom/</code> <code>bom.csv</code> against a value-engineering target of \$25,000 (a hypothetical control target, not a limit), excluding the building shell	Priced equipment list	Over the value-engineering target by \$300: \$25,300 indicative (was \$24,200; +\$1,100 for parts added to make the design buildable, RFE-DDR-003)
R11	Protect workers from fumes	Local exhaust on every melt process with a face velocity of 0.5 m/s or more at the hood; no processing of PVC, PS or unknown plastics; workstation air below the local occupational exposure limits	Hood sizing calculation; later air monitoring	Not verifiable at TRL 3: two enclosing hoods sized for 0.5 m/s (0.57 m ³ /s, 1.00 kW input on a 1.5 kW fan, RFE-DDR-002), filters chosen and limits named; exposure needs air monitoring on a running site
R12	Protect workers from machinery, heat, noise and electricity	Guarded shredder with interlocked hopper and emergency stop; insulated hot surfaces; 30 mA RCD on every circuit near water; noise 85 dB(A) or less over 8 h or hearing protection zones	Safety checklist and design review	At risk: guarding defined; operator exposure 94 dB(A) with a bare shredder and 79 dB(A) in a lined enclosure, based on an assumed sound power
R16	Control input quality	Every delivery sampled at intake; a load whose sampled residue exceeds a set threshold is refused or charged a disposal fee; the threshold keeps R8 within 20 % (new, RFE-DDR-002 item 8)	Mass balance (RFE-CAL-001 section 2); later intake records	Met on paper: threshold 10 % residue by sampled mass (the limit is 10.6 %); the reference input at 20 % residue would fail it

■ Material passport

ID	Requirement	Target	Verification	Status at TRL 3
R13	Travel with every lot that leaves	A passport valid against <code>standards/material-passport-v0.2.schema.json</code> for every product lot, flake bag and export lot (schema named by RFE-DDR-001 item 4; v0.1 is kept)	Schema validation of example records	Met on paper: 3 of 3 example records in <code>standards/examples/</code> are valid
R14	Trace a lot back to its intake batches	Each passport lists its parent batch IDs and the schema version, so a buyer can follow a product back to its collection points	Schema review	Met: v0.2 requires <code>schema_version</code> , and <code>parent_batch_ids</code> for every lot that is not an intake batch
R15	Be quick to fill in	A passport completed in 2 min or less per lot on a phone or a shared laptop at the desk	Timed trial with operators	Not verifiable at TRL 3: needs a form and a timed trial

■ Assumptions

- Reference input: source-separated or picker-sorted post-consumer material, of which about 60 % by mass is target plastic (PET, HDPE, PP). Mixed dumpsite material will yield less.
- One 8 h shift a day, 250 shifts a year, four workers.
- PET is sold as clean flake rather than remanufactured in house, because it must be dried thoroughly and degrades easily in small extruders; HDPE and PP become products. Decided by Amish, 2026-09-25 (RFE-DDR-001 item 3).
- The reference line is plastics only; aluminium is a separate add-on bay with its own safety case. Decided by Amish, 2026-09-25 (RFE-DDR-001 item 1).
- Electrical supply: single-phase, taken as 230 V and 40 A, with staggered heating enforced by an interlock. Decided by Amish, 2026-09-25 (RFE-DDR-001 item 5). Maximum demand 8.75 kW (38.0 A) with the 1.5 kW fume fan (RFE-DDR-002 item 10).
- Sheet press: a derated 5 kW single-phase press is the playbook reference; the published 15 kW press is an option where a 400 V supply exists. Decided by Amish, 2026-09-25 (RFE-DDR-002 item 9).
- Building shell: a rented shed of about 60 m² is preferred; the two-container option is used only where no shed is available, and its side walls are removed only to a structural engineer's design. Decided by Amish, 2026-09-25 (RFE-DDR-002 item 11).
- Costs are indicative USD for new or locally built equipment and exclude land, building, permits and working capital.